

NAVIGATION GENERAL TEST PAPER

1. Given: True course A to B = 250° Distance A to B = 315 NM TAS = 450 kt. W/V = $200^\circ/60$ kt. ETD A = 0650 UTC. What is the ETA at B?

- a) 0810 UTC
- b) 0730 UTC
- ☒ c) 0736 UTC
- d) 0716 UTC

$$\begin{aligned}
 &GS = 409 \\
 &D = 315 \\
 &D = \frac{315}{409} = 0.77 + 0.50 \\
 &= 1.27 \\
 &= 1.27 \times 60 = 76.2 \approx 76
 \end{aligned}$$

2. Given the following: True track: 192° Magnetic variation: $7^\circ E$ Drift angle: 5° left. What is the magnetic heading required to maintain the given track?

- a) 180°
- b) 194°
- c) 204°
- ☒ d) 190°

$$\begin{aligned}
 &C = 192 \\
 &D = 190 \\
 &M = 190 \\
 &Hd = 197 \\
 &Drift = 5^\circ L \\
 &TMG = 192
 \end{aligned}$$

3. 730 FT/MIN equals:

- ☒ a) 3.7 m/sec
- b) 5.2 m/sec
- c) 1.6 m/sec
- d) 2.2 m/sec

730

4. How long will it take to fly 5 NM at a groundspeed of 269 Kt?

- a) 1 MIN 55 SEC
- ☒ b) 1 MIN 07 SEC
- c) 2 MIN 30 SEC
- d) 0 MIN 34 SEC

49

35

①

20 39

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5. Given: Runway direction $230^\circ(T)$, Surface W/V $280^\circ(T)/40$ kt. Calculate the effective cross-wind component?

- a) 38 kt
- ☒ b) 31 kt
- c) 21 kt
- d) 26 kt

$$XW = V \times \sin \theta$$

$$40 \times \sin 50 = 30.64$$

$$230 - 280 = 50$$

6. What is the value of the convergence factor on a Polar Stereographic chart?

- ☒ a) 1.0
- b) 0.866
- c) 0.5
- d) 0.0

$$\begin{aligned} T &= 230 \\ T &= 225 \\ G/S &= 320 \\ TAS &= 360 \end{aligned}$$

7. Given: Magnetic track = 210° , Magnetic HDG = 215° , VAR = $15^\circ E$, TAS = 360 kt, Aircraft flies 64 NM in 12 MIN. Calculate the true W/V?

- ☒ a) $265^\circ/50$ kt
- b) $195^\circ/50$ kt
- c) $235^\circ/50$ kt
- d) $300^\circ/30$ kt

$$D = S \times T$$

$$64 = 12 \times$$

$$= 320 \text{ GS}$$

8. Given: AD = Air distance GD = Ground distance TAS = True Airspeed GS = Ground speed Which of the following is the correct formula to calculate ground distance (GD) gone?

- ☒ a) $GD = (AD \times GS)/TAS$
- b) $GD = (AD - TAS)/TAS$
- c) $GD = AD \times (GS - TAS)/GS$
- ☒ d) $GD = TAS/(GS \times AD)$

9. Given. Distance 'A' to 'B' 2484 NM Mean groundspeed 'out' 420 kt Mean groundspeed 'back' 500 kt Safe endurance 08 HR 30 MIN The distance from 'A' to the Point of Safe Return (PSR) 'A' is:

- a) 1540 NM
- b) 1908 NM
- c) 1736 NM
- d) 1940 NM

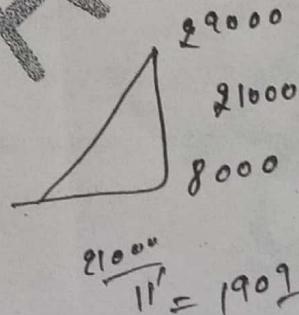
$$\begin{array}{r} 2484 \\ \times 8'30'' \\ \hline 500 \end{array}$$

10. On which of the following chart projections is it NOT possible to represent the north or south poles?

- a) Direct Mercator
- b) Lambert's conformal
- c) Transverse Mercator
- d) Polar stereographic

11. An aircraft at FL290 is required to commence descent when 50 NM from a VOR and to cross that VOR at FL80. Mean GS during descent is 271 kt. What is the minimum rate of descent required?

- a) 2900 FT/MIN
- b) 2000 FT/MIN
- c) 1900 FT/MIN
- d) 1800 FT/MIN



$$\begin{array}{l} D = S \times T \\ 50 = 271 \times T \\ T = 11' \end{array}$$

12. Given: IAS = 235 kt, HDG (T) = 076°W/V = 040/40kt. Calculate the drift angle and GS?

- a) 7R - 204 kt
- b) 7L - 269 kt
- c) 5L - 255 kt
- d) 5R - 207 kt

$$\begin{array}{r} H \quad drift \quad T \\ 76 \quad 75 \quad 8.3 \end{array}$$

$$GS = 204$$

13. Parallels of latitude, except the equator, are:

- a) Rhumb lines

- b) Great circles
- c) both Rhumb lines and Great circles
- d) are neither Rhumb lines nor Great circles

14. Isogonals are lines of equal :

- a) pressure.
- b) compass deviation.
- ☒ c) MAGNETIC VARIATION.
- d) wind velocity.

15. At a specific location, the value of magnetic variation:

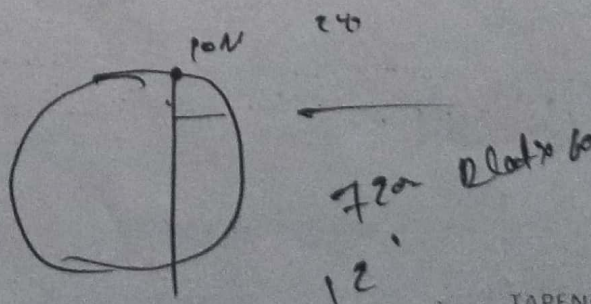
- ☒ a) varies slowly over time
- b) depends on the type of compass installed
- c) depends on the magnetic heading
- d) depends on the true heading

16. Isogonals converge at the

- ☒ a) North and South geographic and magnetic poles
- b) Magnetic equator
- c) North magnetic pole only
- d) North and South magnetic poles only

17. An aircraft at latitude 10° North flies south at a groundspeed of 445 km/HR. What will be its latitude after 3 HR?

- a) $22^{\circ}00'S$
- b) $12^{\circ}15'S$
- ☒ c) $02^{\circ}00'S$
- d) $03^{\circ}50'S$



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* 18. The Earth can be considered as being a magnet with the:

- ☒ a) blue pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface
- b) red pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface
- c) blue pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface
- d) red pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface

19. The horizontal component of the earth's magnetic field:

- ☒ a) is approximately the same at magnetic latitudes 50°N and 50°S
- b) weakens with increasing distance from the magnetic poles
- c) weakens with increasing distance from the nearer magnetic pole
- ☐ d) is approximately the same at all magnetic latitudes less than 60°

20. A flight is to be made from 'A' 49°S , 180°E/W to 'B' 58°S , 180°E/W . The distance in kilometres from 'A' to 'B' is approximately:

- a) 800
- ☒ b) 1000
- c) 540
- d) 804

21. What is the meaning of the term "standard time" ?

- ☒ a) It is the time set by the legal authorities for a country or part of a country
- b) It is the time zone system applicable only in the USA
- c) It is an expression for local mean time
- d) It is another term for UTC

22. Isogonic lines connect positions that have:

- a) the same variation
- ☒ b) 0° variation
- c) the same elevation
- d) the same angle of magnetic dip

23. The circumference of the parallel of latitude at 60°N is approximately:

- a) 10 800 KM
- ☒ b) 10 800 NM
- c) 20 000 NM
- d) 34 641 NM

24. At what approximate latitude is the length of one minute of arc along a meridian equal to one NM (1852 m) correct?

- a) 90°
- b) 0°
- ☒ c) 45°
- d) 30°

25. The duration of civil twilight is the time:

- ☒ a) between sunset and when the centre of the sun is 6° below the true horizon
- b) agreed by the international aeronautical authorities which is 12 minutes
- c) needed by the sun to move from the apparent height of 0° to the apparent height of 6°
- d) between sunset and when the centre of the sun is 12° below the true horizon

26. What is the longitude of a position 6 NM to the east of $58^\circ 42'\text{N } 094^\circ 00'\text{W}$?

$$PL = \Delta \log \times 60 \times \cos \text{Lat}$$

$$6 = \Delta \log \times 60 \times \cos 58' 49''$$

$$\Delta \log = 11''$$

- A) 093°48.5'W
- B) 093°54.0'W
- C) 093°53.1'W
- D) 094°12.0'W

27. At what approximate date is the earth closest to the sun (perihelion)?

- a) Beginning of JULY
- b) End of March
- C) Beginning of JANUARY
- d) End of June

28. An aircraft flies the following rhumb line tracks and distances from position 04°00'N 030°00'W : 600 NM South, then 600 NM East, then 600 NM North, then 600 NM West. The final position of the aircraft is:

- a) 04°00'N 029°58'W
- b) 04°00'N 030°02'W
- c) 04°00'N 030°00'W
- d) 03°58'N 030°02'W

29. The purpose of compass check swing is to:

Deviation

- a) measure the angle between Magnetic North and Compass North
- b) cancel out the vertical component of the earth's magnetic field
- c) cancel out the horizontal component of the earth's magnetic field.
- d) cancel out the effects of the magnetic fields found on board the aeroplane

30. On a Lambert conformal conic chart, the distance between parallels of latitude spaced the same number of degrees apart :

- a) CONSTANT THROUGHOUT CHART
- b) is constant between, and expands outside, the standard parallels

35 An air

- c) expands between, and reduces outside, the standard parallels
a) reduces between, and expands outside, the standard parallels

31. Fuel flow per HR is 22 US-GAL, total fuel on board is 83 IMP GAL. What is the endurance?

- a) 4 HR 12 MIN
b) 4 HR 32 MIN
c) 3 HR 53 MIN
d) 2 HR 15 MIN

Handwritten calculations for Q31:
 $83 \times 1.2 = 99.6$
 $\frac{99.6}{22} = 4.527$
18.3 2' 4' 31'

32. What is the ratio between the litre and the US-GAL ?

- a) 1 US-GAL equals 3.78 litres
b) 1 litre equals 3.78 US-GAL
c) 1 US-GAL equals 4.55 litres
d) 1 litre equals 4.55 US-GAL

33. 265 US-GAL equals? (Specific gravity 0.80)

- a) 895 kg
b) 862 kg
c) 803 kg
d) 940 kg

Handwritten calculation for Q33:
 $265 \times 3.7854 = 1002.3$
 $\times 0.8$
 $= 802$

34. An aeroplane is flying at TAS 180 kt on a track of 090°. The W/V is 045° / 50kt. How far can the aeroplane fly out from its base and return in one hour?

- a) 85 NM
b) 88 NM
c) 56 NM
d) 176 NM

Handwritten diagram and calculations for Q34:
Diagram: A triangle with a horizontal base and two slanted sides. The left slanted side is labeled 211.85, the right slanted side is labeled 141.14.
Calculation:
 $\frac{180}{0.44} = 409$
 $\frac{409}{2} = 204.5$
 ≈ 205

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35. An aircraft is ""homing"" to a radio beacon whilst maintaining a relative bearing of zero. If the magnetic heading decreases, the aircraft is experiencing :

- a) right drift
- ☒ b) left drift
- c) zero drift
- d) a wind from the west

* 36. Given : Compass heading 270° Deviation $2^\circ W$ Variation $30^\circ E$ Relative bearing 316° What is the QDR?

- a) 046°
- b) 226°
- ☒ c) 044°
- d) 224°

Handwritten calculations for Q36:
 $C = 270$
 $D = 2W$
 $M = 288$
 $QDR = 044$
 $MH + RB = 224$

37. The angular difference, on a Lambert conformal conic chart, between the arrival and departure track is equal to:

- ☒ a) map convergence
- b) earth convergence
- c) conversion angle
- d) difference in longitude

* 38. What is the chart distance between longitudes $179^\circ E$ and $175^\circ W$ on a direct Mercator chart with a scale of 1 : 5 000 000 at the equator?

- a) 106 mm
- ☒ b) 133 mm
- c) 167 mm
- d) 72 mm

$$\frac{1}{5000000} = \frac{179 + 175}{\text{Distance}}$$

39. The constant of the cone, on a Lambert chart where the convergence angle between longitudes 010°E and 030°W is 30°, is

- a) 0.40
- ☒ b) 0.75
- c) 0.50
- d) 0.64

30 = 0.69° 30 min 10 sec

30 = 40

30 = 0.75
40

40. 5 HR 20 MIN 20 SEC corresponds to a longitude difference of:

- ☒ a) 80°05'
- b) 81°10'
- c) 75°00'
- d) 78°45'

41. Given: Maximum allowable take-off mass 64400 kg, Maximum landing mass 56200 kg, Maximum zero fuel mass 53000 kg, Dry operating mass 35500 kg, Traffic load 14500 kg, Trip fuel 4900 kg, Take-off fuel 7400 kg Find: Maximum additional load

- A) 4000 kg
- ☒ B) 3000 kg
- C) 7000 kg
- D) 5600 kg

MTOM

MLM

MZFM

64400

56200

53000

4900

7400

61100

60400

41. On a ATC flight plan, to indicate that you will overfly the way-point ROMEO at 120 kt at flight level 085, you will write :

- A) ROMEO / NO120 F085
- B) ROMEO / K0120 FL085
- C) ROMEO / FL085 NO120
- ☒ D) ROMEO / F085 NO120

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42. Given: Dry operating mass (DOM) = 33510 kg Load = 7600 kg Final reserve fuel = 983 kg Alternate fuel = 1100 kg Contingency fuel 102 kg The estimated landing mass at alternate should be :

- A) 42495 kg.
- ☒ B) 42195 kg.
- C) 42210 kg.
- D) 42312 kg.

$$33510 + 7600 + 983 + 102$$

43. A "current flight plan" is a :

- ☒ A) filed flight plan with amendments and clearance included.
- B) flight plan with the correct time of departure.
- C) flight plan in the course of which radio communication should be practised between aeroplane and ATC

44. In the ATS flight plan Item 15 (Cruising speed), when not expressed as a Mach number, cruising speed is expressed as :

- ☒ A) CAS
- B) IAS
- ☒ C) TAS
- ☐ D) Groundspeed

45. An aircraft has a maximum certificated take-off mass of 137000 kg but is operating at take-off mass 135000 kg. In Item 9 of the ATS flight plan its wake turbulence category is :

- ☒ A) heavy "H"
- B) heavy/medium "H/M"
- C) medium "M"
- D) medium plus "

46. A descent is planned from 7500 ft MSL so as to arrive at 1000 ft MSL 6 NM from a VORTAC. With a GS of 156 kts and a rate of descent of 800 ft/min. The distance from the VORTAC when descent is started is :

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46. ☒ A) 27,1 NM
B) 15,0 NM
C) 11,7 NM
D) 30,2 NM

47. The navigation plan reads: Trip fuel: 100 kg Flight time: 1h35min Taxi fuel: 3 kg Block fuel: 181 kg The endurance on the ICAO flight plan should read:

- A) 2h 04min
B) 1h 35min
☒ C) 2h 49min
D) 2h 52min

48. If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

- ☒ A) ZZZZ
B) AAAA
C) XXXX

49. An aircraft at position $60^{\circ}\text{N } 005^{\circ}\text{W}$ tracks $090^{\circ}(\text{T})$ for 315 km. On completion of the flight the longitude will be:

- ☒ a) $000^{\circ}40'\text{E}$
b) $005^{\circ}15'\text{E}$
c) $002^{\circ}10'\text{W}$
d) $000^{\circ}15'\text{E}$

50. The 'departure' between positions $60^{\circ}\text{N } 160^{\circ}\text{E}$ and $60^{\circ}\text{N } 'x'$ is 900 NM. What is the longitude of 'x'?

- ☒ A) 175°W

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- b) 140°W
- c) 145°E
- ☒ d) 170°E

51. The north and south magnetic poles are the only positions on the earth's surface where:

- ☒ a) a freely suspended compass needle will stand vertical
- b) isogonals converge
- c) a freely suspended compass needle will stand horizontal
- ☒ d) the value of magnetic variation equals 90°

52. Given: True HDG = 145°, TAS = 240 kt, Track (T) = 150°, GS = 210 kt. Calculate the W/V?

- a) 195/35kt
- b) 360/35kt
- ☒ c) 115/35kt
- d) 295/35kt

53. Given: Distance 'A' to 'B' 2346 NM Groundspeed 'out' 365 kt Groundspeed 'back' 480 kt Safe endurance 8 HR 30 MIN The time from 'A' to the Point of Safe Return (PSR) 'A' is:

- a) 239 MIN
- b) 219 MIN
- c) 197 MIN
- ☒ d) 290 MIN

$$\frac{2346}{\frac{480 \times 8.5}{365 + 480}} = 290$$

54. An aircraft basic empty mass is 3000 kg. The maximum take-off, landing, and zero-fuel mass are identical, at 5200 kg. Ramp fuel is 650 kg, the taxi fuel is 50 kg. The payload available is:

- a) 1 550 kg
- ☒ b) 1 600 kg
- c) 2 200 kg

d) 2 150 kg

55. Given: Dry Operating Mass = 29 800 kg Maximum Take-Off Mass = 52 400 kg Maximum Zero-Fuel Mass = 43 100 kg Maximum Landing Mass = 46 700 kg Trip fuel = 4 000 kg Fuel quantity at brakes release = 8 000 kg The maximum traffic load is:

- a) 13 300 kg
- ☒ b) 12 900 kg
- c) 9 300 kg
- d) 14 600 kg

56. Dry Operating Mass is the mass of the aeroplane less

- ☒ a) usable fuel and traffic load.
- b) usable fuel.
- c) traffic load, potable water and lavatory chemicals.
- ☒ d) usable fuel, potable water and lavatory chemicals.

57. The Dry Operating Mass of a helicopter is the total mass of a helicopter :

- ☒ a) ready for a specific operation including the crew and traffic load, not including the usable fuel
- b) excluding the crew but including specific equipments for the mission and not including the usable fuel
- ☒ c) including the crew, the fuel and the specific equipments for the mission but excluding payload
- d) including the crew, the usable fuel and the specific equipments for the mission and payload

58. The maximum taxi (ramp) mass is governed by :

- ☒ a) structural considerations.
- b) tyre speed and temperature limitations
- c) bearing strength of the taxiway pavement.

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d) taxi distance to take - off point

59. The maximum certificated taxi (or ramp) mass is that mass to which an aeroplane may be loaded prior to engine start. It is :

- a) A fixed value which is listed in the Flight Manual.
- b) A value which varies with airfield temperature and altitude. Corrections are listed in the Flight Manual.
- c) A value which varies only with airfield altitude. Standard corrections are listed in the Flight Manual.
- d) A value which is only affected by the outside air temperature. Corrections are calculated from data given in the Flight Manual.

60. A great circle track joins position A ($59^{\circ}\text{S } 141^{\circ}\text{W}$) and B ($61^{\circ}\text{S } 148^{\circ}\text{W}$). What is the difference between the great circle track at A and B?

- a) It increases by 3°
- b) It decreases by 6°
- c) It increases by 6°
- d) It decreases by 3°

CON = DEGA CHIN
E W 60
N I D
S D I

61. A negative (westerly) magnetic variation signifies that :

- a) True North is East of Magnetic North
- b) True North is West of Magnetic North
- c) Compass North is East of Magnetic North
- d) Compass North is West of Magnetic North

62. The nominal scale of a Lambert conformal conic chart is the:

- a) scale at the standard parallels

- b) mean scale between pole and equator
- c) mean scale between the parallels of the secant cone
- d) scale at the equator

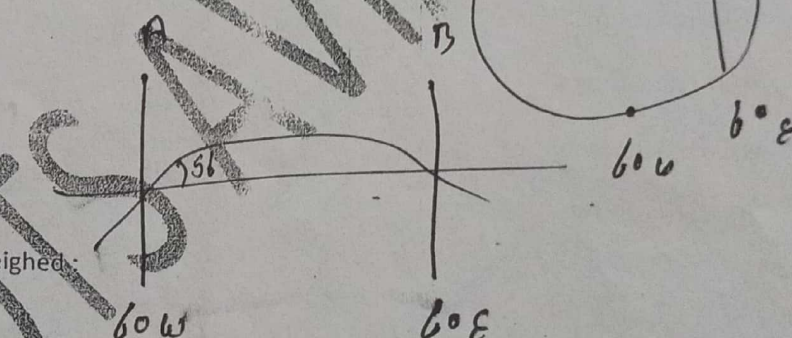
63. On a Direct Mercator chart, a rhumb line appears as a:

- ☒ a) straight line
- b) small circle concave to the nearer pole
- c) spiral curve
- d) curve convex to the nearer pole

64. On a Polar Stereographic chart, the initial great circle course from A $70^{\circ}\text{N } 060^{\circ}\text{W}$ to B $70^{\circ}\text{N } 060^{\circ}\text{E}$ is approximately:

- a) 210° (T)
- b) 330° (T)
- c) 150° (T)
- ☒ d) 030° (T)

210° to 1°N



65. An aeroplane may be weighed:

- a.) in quiet parking area
- b.) at specified weighing location
- ☒ c.) in an enclosed, non airconditioned hanger
- d.) in an area of airside set aside of maintenance

66. For a repetitive flight plan to be used, flights must take place on regular basis on atleast:

- ☒ a) 10 occasions
- b.) 20 occasions
- c.) 25 occasions
- d.) 15 occasions

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67.) The fuel burn of an aircraft turbine engine is 220L/H with a fuel density of 0.8. if the density is 0.75, the fuel burn will be:

a.) 220L/H

☒ b.) 235L/H

c.) 176 L/H

68.) Minimum planned take-off fuel is 160kg (30% total reserve fuel is included). Groundspeed on the trip is constant. When the aeroplane has done half the distance the remaining fuel is 70 kg. Is diversion to a nearby alternate necessary?

a.) Diversion to nearby alternate is not necessary, because the reserve has not been used completely.

☒ b.) Diversion to nearby alternate is necessary, because the remaining fuel is not sufficient.

c.) Diversion to nearby alternate is necessary unless pilot decides to continue on his own responsibility.

69. How many hours in advance of departure time should a flight plan be filed in the case of flights into areas subject to air traffic flow management (ATFM)?

☒ a) 3:00 hours.

b) 1:00 hour.

c) 0:30 hours.

d) 0:10 hours.

70. For a flight plan filed before the flight, the indicated time of departure is:

☒ a) the estimated off-block time

b) the time at which the flight plan is filed.

c) the time of take-off.

d) the time overhead the first reporting point after take-off.

71. The equivalent of 70 m/sec is approximately:

- 57 NN 15

19

- (

2222

- c D m VT
 215 15E 230
- d dit + TM 6
- 30

195°/50 kt
235°/50 kt
300°/30 kt

4. A straight line drawn on a chart measures 4
1 : 3 000 000
1 : 6 000 000
1 : 5 000 000

- $$\frac{4.63}{150} = 0.000000$$

a) 81 NM

Scanned with CamScanner

- b) 57 NM
c) 70 NM
d) 86 NM

$$\frac{1}{70} \times \frac{\cos B 30}{\cos B 45}$$

76. Given: FL 120, OAT is ISA standard, CAS is 200 kt, Track is $222^\circ(M)$, Heading is $215^\circ(M)$, Variation is $15^\circ W$ Time to fly 105 NM is 21 MIN. What is the W/V?

- a) $050^\circ(T) / 70$ kt.
b) $040^\circ(T) / 105$ kt.
c) $055^\circ(T) / 105$ kt.
d) $065^\circ(T) / 70$ kt.

$$CAS = 300$$

$$\begin{aligned} T - Hdg &= 200 \\ T - T &= 207 \\ GS &= 300 \\ TRS &= 238 \end{aligned}$$

77. Given: TAS = 200 kt, Track (T) = 073° , W/V = 210/20 kt. Calculate the HDG ($^\circ T$) and GS?

- a) 077 - 214 kt
b) 079 - 211 kt
c) 075 - 213 kt
d) 077 - 210 kt

78. A Lambert conformal conic chart has a constant of the cone of 0.80. A straight line course drawn on this chart from A ($53^\circ N 004^\circ W$) to B is 080° at A, course at B is $092^\circ(T)$. What is the longitude of B?

- a) $018^\circ E$
b) $009^\circ 36' E$
c) $011^\circ E$
d) $019^\circ E$

$$0.80 = 12$$

$$9^\circ 36'$$

$$080 - 92 = 12$$

$$12 = \log \sin 110^\circ$$

$$12 = D \times 0.8 - 15 - 11 =$$

79. On a Transverse Mercator chart, scale is exactly correct along the:

- a) meridian of tangency
b) Equator, parallel of origin and prime vertical

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- c) datum meridian and meridian perpendicular to it
- d) prime meridian and the equator

80. Transverse Mercator projections are used for:

- ☒ a) maps of large north/south extent
- b) maps of large east/west extent in equatorial areas
- c) radio navigation charts in equatorial areas
- d) plotting charts in equatorial areas

81. The Dry Operating Mass of an aircraft is 2 000 kg. The maximum take-off mass, landing and zero fuel mass are identical at 3500 kg. The block fuel mass is 550 kg, and the taxi fuel mass is 50 kg. The available mass of payload is:

- ☒ a) 1 500 kg
- b) 950 kg
- ☒ c) 1 000 kg
- d) 1 450 kg

82. What is the value of the magnetic dip at the magnetic south pole ?

- a) 60°
- b) 45°
- ☒ c) 90°
- d) 0°

83. The value of magnetic variation:

- ☒ a) has a maximum of 180°
- b) must be 0° at the magnetic equator
- c) varies between a maximum of 45° East and 45° West

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d) cannot exceed 90°

84. What is the local mean time, position $65^\circ 25' N$ $123^\circ 45' W$ at 2200 UTC?

- a) 1345
- b) 2200
- c) 615
- d) 815

85. If an aeroplane was to circle around the Earth following parallel $60^\circ N$ at a ground speed of 480 kt. In order to circle around the Earth along the equator in the same amount of time, it should fly at a ground speed of:

- a) 460 kt
- b) 240 kt
- c) 960 kt
- d) 480 kt

* 86. Conversion of fuel volume to mass

- a) may be done by using standard fuel density values as specified in the Operations Manual, if the actual fuel density is not known.
- b) may be done by using standard fuel density values as specified in JAR - OPS 1.
- c) must be done by using actual measured fuel density values.

87. The actual 'Zero Fuel Mass' is equal to the:

- a) Dry Operating Mass plus the traffic load.

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- b) Operating Mass plus all the traffic load.
- c) Basic Empty Mass plus the fuel loaded.
- d) Actual Landing Mass plus trip fuel.

88. The atmospheric pressure at FL 70 in a "standard + 10" atmosphere is:

☒ a) 781.85 hPa.

b) 942.13 hPa.

c) 1 013.25 hPa.

d) 644.41 hPa.

$$QNE 1013.25$$

$$\frac{7000}{30} = 233$$

$$= 1013.25 - 233 = 780.25 + 10$$

$$= 789.9 \text{ approx.}$$

89. At 60S on a mercator chart, chart convergence is:

☒ a.) lesser than earth convergency

b.) correct

c.) greater than earth convergency

d.) equal to $ch \text{ long} \times 0.8666$

90. what is the approximate compression of earth?

a.) 3%

b.) 0.03%

☒ c.) 0.3%

d.) 1/3000

91. An aircraft is maintaining 5.2% gradient on a flat terrain. Its height at 7nm from the runway is approximately?

a.) 3640 feet

b.) 1890 feet

c.) 2210 feet

d.) 680 feet

$$\% = \frac{H + 100}{2}$$

$$S.Z = \frac{H}{7} \times 100$$
$$= 2213$$

92. A non-perspective chart:

a.) is produced directly from a light projection of reduced earth.

b.) cannot be used for navigation.

c.) is produced by mathematically adjusting a light projection of the reduced earth.

d.) is used for stereographic projection.

93. At 0422 an aircraft at FL370, GS 320kt, is on the direct track to VOR 185Nm distant. the aircraft is required to cross vor at FL 80. For a mean rate of descent

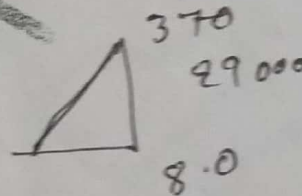
of 1800ft/MIN at mean GS of 232 kt, the latest time at which to commence descent is?

a.) 0451

b.) 0454

c.) 0445

d.) 0448



18.11

$$= D = \frac{S \times T}{180}$$
$$189 = \frac{320 \times T}{180}$$

94. The sun rises at 50N and 25W on the 25th jan. at 0654 utc. on the same day what time does it rise at 50N and 040E?

a.) 0254 UTC

b.) 0154 UTC

c.) 0234 UTC

d.) 0514 UTC

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95. If the PNR is calculated 880NM with 10,000kg of fuel available, the distance to the PNR with 11,000 kg fuel available, other factors being, will be-

a.) 920Nm

b.) 968Nm

c.) 960nm

$$\frac{10,000}{11,000} \times \frac{880}{10,000} \times 11,000 = 968$$

96. Cruising at FL 390, M0.84 is found to give a TAS of 499kt. the ISA deviation at this level will be:

a.) -17

b.) -19

c.) +17

d.) +19

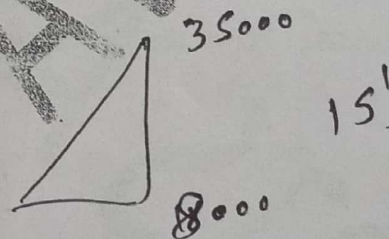
97. An aircraft at FL350 is required to descend to cross a DME facility at FL80. Maximum rate of descent is 1800 FT/MIN and mean GS for descent is 276 kt. The minimum range from the DME at which descent should start is:

a) 69 NM

b) 79 NM

c) 49 NM

d) 59 NM



98. Given TAS = 190 kt, True HDG = 085°, W/V = 110°(T)/50kt. Calculate the drift angle and GS?

a) 4°L - 145 kt

b) 7°L - 156 kt

c) 4°L - 168 kt

d) 7°L - 146 kt

Handwritten calculations for Q98:
 Hd 085
 drift 75
 GS 146
 W/V 110/50
 S

99. Given: True HDG = 307°, TAS = 230 kt, Track (T) = 313°, GS = 210 kt. Calculate the W/V?

a) 260/30kt

- b) 257/35kt
- c) 255/25kt
- d) 265/30kt

100. Given: true track 070° variation 30°W deviation +1° drift 10°R Calculate the compass heading?

- a) 091°
- ☒ b) 089°
- c) 100°
- d) 101°

Handwritten notes and calculations:

C 89
 Hd 60
 D 30 1E 90
 drift 10S
 TMH 40
 300 60

KNOTS